Eigenvalues and Eigenvectors

		vector space	
7, 8, 9	Jan 23, 25, 30	Matrix as Linear Transformation	MatrixAsLinearMappings.pdf Matlab file: example3.m Matlab (rotation animation): rotation1.m
10	Feb 1	Eigenvalues	Matlab file: eigen_demo1.m

Eigenvalues and Eigenvectors

Recall: A : 2×2 matrix

$$\begin{bmatrix} x_i \\ y_i \end{bmatrix} \longmapsto \begin{bmatrix} a_i \\ b_i \end{bmatrix} = A \begin{bmatrix} x_i \\ y_i \end{bmatrix}$$

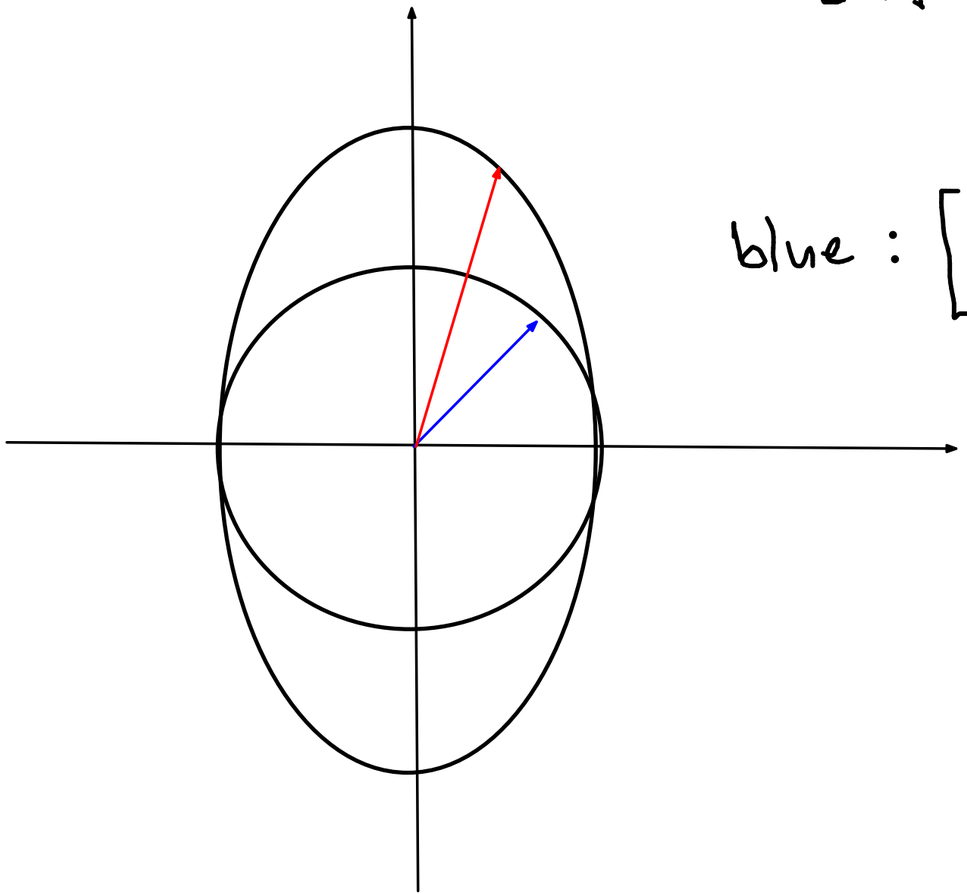
A maps $\begin{bmatrix} x_i \\ y_i \end{bmatrix}$ to the vector $\begin{bmatrix} a_i \\ b_i \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 2y \end{bmatrix}$$

red : $A \begin{bmatrix} x_i \\ y_i \end{bmatrix}$

blue : $\begin{bmatrix} x_i \\ y_i \end{bmatrix}$



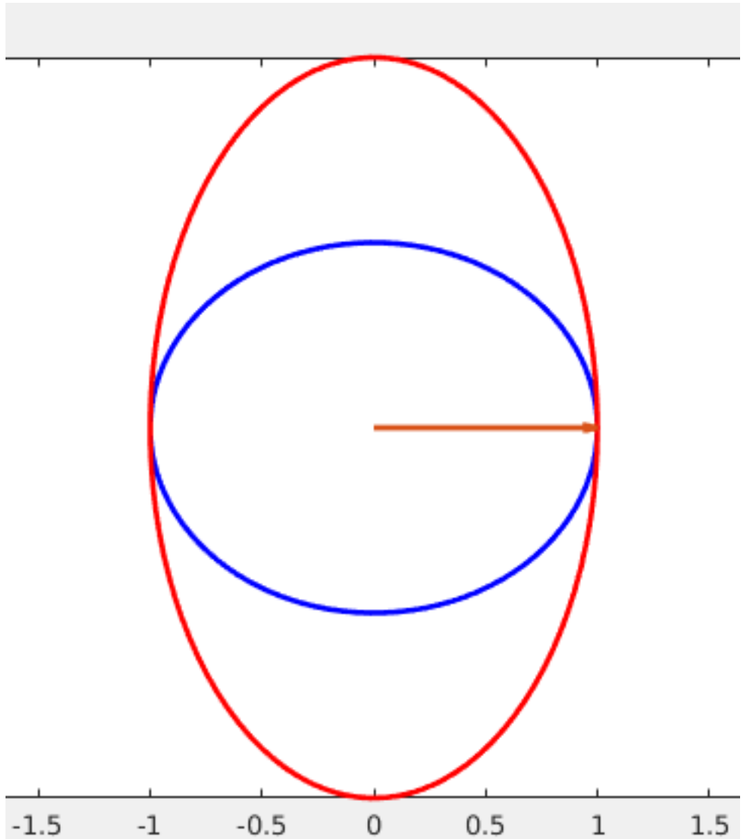
Eigenvalues and Eigenvectors

→ A is 2x2 matrix

→ A is mapping from 2d space to 2d space

→ Eigenvalues : measures the amount of distortion A generates

→ Eigenvectors : direction of distortion



$$A : \{x^2 + y^2 = 1\} \to \left\{x^2 + \frac{y^2}{4} = 1\right\}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

one eigenvector: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ & eigenvalue is 1.

one eigenvector: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ & eigenvalue is 2.

```
>> [v ,d] = eig(A)

v =

     1     0
     0     1

d =

     1     0
     0     2
```

eigenvectors

diagonal matrix.
diagonal entries are eigenvalues

eigenvector for eigenvalue 1 is $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

& eigenvector for eigenvalue 2 is $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

1 Determination of Eigenvalues and Eigenvectors

$$A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$$

$A v_1 = -6 v_1$

$A v_2 = -1 v_2$

In general:

eigenvalue, eigenvector

λ $\vec{v}$

$$A\vec{v} = \lambda \vec{v}$$

$\downarrow$ 2 equations & 3 unknowns

```
A =

   -5    2
    2   -2

>> [v, d] = eig(A)

v =

   -0.8944   -0.4472
    0.4472   -0.8944

d =

   -6    0
    0   -1
```

v_1 v_2

-6 -1

$-6, -1$] eigenvalues

Theorem: if A is a $m \times m$ square matrix & $\vec{v}$ is a $m \times 1$ non zero vector such that

$$A\vec{v} = \vec{0},$$

then, A is not invertible

$$\Leftrightarrow \det(A) = 0.$$

$$A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$$

we want to find eigenvalues (λ) & eigenvectors (v)

<u>note:</u> eigenvector can not be zero.

$$v = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$Av = \lambda v$$

$$\begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - \lambda \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} -5-\lambda & 2 \\ 2 & -2-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

by definition of eigenvector, $\begin{bmatrix} x \\ y \end{bmatrix} \neq 0$.

$$\begin{vmatrix} -5-\lambda & 2 \\ 2 & -2-\lambda \end{vmatrix} = 0$$

defining equ for eigenvalues

$$(-5-\lambda)(-2-\lambda) - 4 = 0$$

$$\lambda = -6, -1$$

in general: eigenvalues of A satisfy the eq^n

$$\det(A - \lambda I) = 0$$

$A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$ We just found eigenvalues $\lambda = -6, -1$, by solving $|A - \lambda I| = 0$

Lets find eigenvectors for $\lambda = -6$

let $v = \begin{bmatrix} x \\ y \end{bmatrix}$ is an eigenvector for $\lambda = -6$

by definition $Av = -6v$

$$\begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = -6 \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

↳ not invertible

↳ solve using Gaussian elimination

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \end{bmatrix}$$

$$\Big\downarrow \; R_2 \to R_2 - 2R_1$$

$$\begin{bmatrix} \boxed{1} & 2 & 0 \\ \boxed{0 \quad 0} & \boxed{0} \end{bmatrix}$$

infinitely many solutions

x : pivot variable

y : free variable

$$x + 2y = 0$$

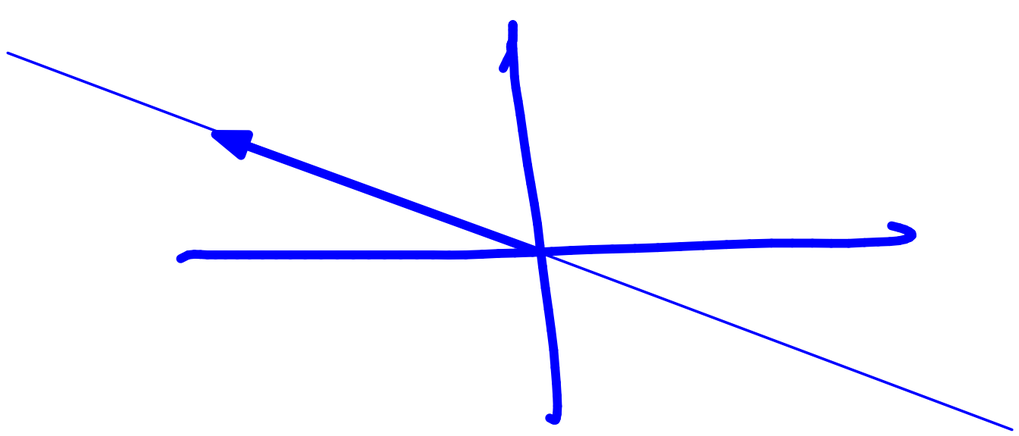
$$x = -2y$$

eigenvectors

$$v = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2y \\ y \end{bmatrix} = y\begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

all non zero multiple of $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$ is an eigenvector for A $\lambda = -6$

$$A v = -6 \vec{v}$$



$$A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$$

We just found eigenvalues $\lambda = -6, -1$, by solving $|A - \lambda I| = 0$

Lets find eigenvectors for $\lambda = -1$

$$v = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \text{then } 2x - y = 0$$

$$= \begin{bmatrix} y/2 \\ y \end{bmatrix} = y \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}$$

2 Multiple Eigenvalues

manually first, then verify your answer on matlab.

Find the eigenvalues and eigenvectors of

$$\mathbf{A} = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

4 Real Matrices with Complex Eigenvalues and Eigenvectors

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$